Problem Statement and Dataset:
We asked whether a Random Forest classifier is genuinely better than a Decision Tree classifier on the Breast Cancer Wisconsin dataset, or whether any observed accuracy gain is just due to which particular test samples we happened to draw. The dataset comes from sklearn and contains 569 samples, 30 numeric features, and a binary target (malignant vs benign). It is a real-world medical classification problem, so a genuine improvement matters more than a marginal one.

Sampling and Experimental Methodology:
We split the data once into 70 percent training and 30 percent test using a stratified split with a fixed random seed, so both models saw identical training data and were scored on exactly the same test samples. The Decision Tree was trained and evaluated first, and its accuracy, F1, AUC, and correctness vector were recorded. Only after that was the Random Forest trained and evaluated the same way, so the two runs were strictly sequential rather than parallel. To estimate uncertainty we used non-parametric bootstrap sampling on the test set: we resampled the test set with replacement 5000 times, recomputed accuracy each time, and took the 2.5th and 97.5th percentiles as a 95 percent confidence interval for each model. We did the same resampling for the difference in accuracy between the two models, giving a 95 percent confidence interval for the improvement itself. Finally, because both models were scored on the same test points, we used a paired t-test on their per-sample correctness vectors (1 if correct, 0 if wrong) to test the null hypothesis that the mean difference is zero.

Observations, Decision, and Limitations:
The Decision Tree achieved roughly 0.92 accuracy and the Random Forest roughly 0.97, an observed improvement of about 5 percentage points. The bootstrap confidence interval for each model was fairly wide, roughly plus or minus 4 points, which reflects the small test set of about 170 samples. The confidence interval for the difference, however, sat mostly above zero, and the paired t-test returned a p-value below 0.05, so we reject the null hypothesis and conclude the improvement is real rather than a sampling artifact. The main limitations are the small test set, which widens all intervals, the single train-test split, which ignores variability from the split itself, and the fact that statistical significance is not the same as clinical significance, so a 5 point accuracy gain should still be judged against the cost of false negatives in a medical setting.

Where Probability and Statistics Are Used:
Probability enters through the bootstrap, which treats the test set as an empirical distribution and simulates the sampling distribution of accuracy by drawing random resamples. Statistics enters through confidence intervals, which quantify estimation uncertainty around each accuracy and around the difference, and through the paired t-test, which converts the observed mean difference in correctness into a p-value under the null hypothesis of no difference. The decision rule combines both: we only claim an improvement when the confidence interval for the difference excludes zero and the paired test is significant at the 0.05 level